

bonds, the risks are inflation, interest-rate increases, taxes, and a potential default by the issuer.

Portfolio risk cannot be eliminated although it can be partially controlled with an asset allocation strategy. Combining different investment types, each with its own unique risk and return characteristics, into one portfolio creates a unique risk and return trade-off in the portfolio. This is similar to making bread from flour, yeast, and water. The combined product has different characteristic from its ingredients.

A well-designed portfolio lowers overall risk through diversification, and this eventually results in a return for the portfolio that's higher than the weighted average return of the individual investments in the portfolio. This risk reduction phenomenon does not happen every year, but it does happen over several years if you are a disciplined investor. Once you grasp the mechanics behind asset allocation and accumulate information on different types of investments, you will be ready to design a portfolio that has an expected return and an acceptable level of risk for your needs.

THE MYTHICAL RISK-FREE INVESTMENT

The lowest-risk investment in the U.S. financial markets is a U.S. Treasury bill (T-bill), a government-guaranteed investment that matures in one year or less. T-bills are sold at a discount from face value and don't pay interest before maturity. The interest is the difference between the purchase price of the bill and its face value paid at maturity.

The U.S. Treasury issues T-bills weekly, and the interest rate is set by an auction system. The current T-bill rate is a good proxy for the interest an investor will earn in a money market fund because T-bills are frequently purchased in money market funds.

T-bills are often called a "risk-free" investment in the financial world because of their short maturity and government guaranteed return. However, *risk-free* may be an inappropriate choice of words. T-bills do have a reliable positive return; however, that return is subject to the corrosive effects of taxes and inflation.

[Figure 2-1](#) highlights the year-over-year T-bill return minus 25 percent income tax and the inflation rate. There have been many years when the rate of return on T-bills has not kept pace with the inflation rate after taxes. Investors in T-bills and money market funds are losing purchasing power in